

MULTI-STAGE GENAI PROMPT WORKFLOW

AI Interview Preparation Assistant

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Programming Language: Python

1. Scenario Selected

The selected scenario is an AI Interview Preparation Assistant. The system is intended to help a student prepare for placement interviews by examining the student's current skills, identifying areas that need more practice, suggesting a preparation sequence, and generating practice questions. The workflow demonstrates how multiple GenAI prompts can work together instead of depending on one large prompt.

2. Objective

Objective	Description
Skill assessment	Interpret a student's current preparation level from topic-wise performance.
Gap detection	Highlight subjects where additional revision is required.
Preparation planning	Create a practical sequence for interview preparation.
Question generation	Produce questions according to the learner's current level.
Feedback	Explain common mistakes and suggest the next area to practise.

3. Sample Candidate Information

Area	Score	Attempts	Interpretation
Python Fundamentals	82%	20	Good foundation
SQL	63%	16	Needs revision
Data Structures	57%	18	Needs practice
Computer Networks	74%	12	Moderate
Communication / HR	68%	10	Can be improved

4. Stage 1 – Candidate Analysis

Prompt used: Analyze the candidate profile and classify the preparation areas into strong, moderate, and priority categories. Explain which areas should be revised first and create a short interview-preparation strategy. Give priority to measurable performance rather than making assumptions about the candidate.

Intermediate AI response: Python is currently the strongest technical area. Data Structures and SQL should receive more attention because their scores are comparatively lower. Computer Networks can be maintained with revision, while communication practice should be included regularly because interview performance depends on both technical knowledge and the ability to explain answers clearly.

5. Stage 2 – Preparation Plan Generation

Prompt used: Using the candidate analysis, construct a five-day interview preparation plan. Every day must include one technical focus, one practical task, a small set of interview questions, and a self-check activity. Avoid placing all difficult topics on the same day and include revision of previously studied areas.

Generated Preparation Plan

Day	Main Focus	Practice Activity	Self-check
1	Data Structures	Arrays, strings and complexity questions	10-question quiz

2	SQL	Queries, joins and aggregation problems	Write 5 queries
3	Python	Functions, collections and debugging	Solve 2 coding tasks
4	Networks + HR	Networking concepts and HR responses	Mock interview
5	Full Revision	Mixed technical question set	Timed assessment

6. Stage 3 – Evaluation and Refinement

Review prompt: Act as a placement-training reviewer. Check whether the proposed schedule is balanced, whether weak areas receive enough attention, whether the activities can realistically be completed in one day, and whether the plan contains a way to measure improvement. List problems first and then provide a revised plan.

Review findings: The original schedule should avoid spending an entire day on passive reading. SQL and Data Structures require active problem solving. Communication practice should not be isolated from technical preparation because the student must explain technical answers during an interview. The final day should measure both accuracy and response time.

7. Refined Workflow

Candidate Data → Skill Analysis → Priority Identification → Preparation Schedule → Question Generation → Practice → Performance Review → Updated Preparation Plan

8. Main GenAI Components

Component	Role in the workflow
Profile analysis	Converts performance information into preparation priorities.
Adaptive question generation	Creates questions suited to the candidate's current level.
Mock interview support	Provides a structured set of technical and HR questions.
Answer evaluation	Checks an answer for correctness, clarity and missing points.
Plan refinement	Changes the next preparation cycle based on recent performance.

9. Expected Output

The final system should provide the learner with a prioritized preparation list, a short-term study schedule, topic-specific interview questions, feedback on practice performance, and recommendations for the next revision cycle. The design keeps the workflow practical for a college-level project while demonstrating chained prompting and self-evaluation.

10. Conclusion

This assignment demonstrates a three-stage GenAI workflow for interview preparation. The first stage interprets candidate performance, the second converts that interpretation into an actionable plan, and the third evaluates the plan before it is finalized. Because new practice results can be fed back into the workflow, the preparation process can become progressively more targeted over time.